

AI Assisted Coding Assignment- 10.2

2303A51886 : 30 SAI SPURTHI. B 24.02.2026

Lab 10 – Code Review and Quality: Using AI to Improve Code Quality and Readability

Lab Objectives

- Use AI for automated code review and quality enhancement.
- Identify and fix syntax, logical, performance, and security issues in Python code.
- Improve readability and maintainability through structured refactoring and comments.
- Apply prompt engineering for targeted improvements.
- Evaluate AI-generated suggestions against PEP 8 standards and software engineering best practices

Task Description -1(Error Detection and Correction)

Task:

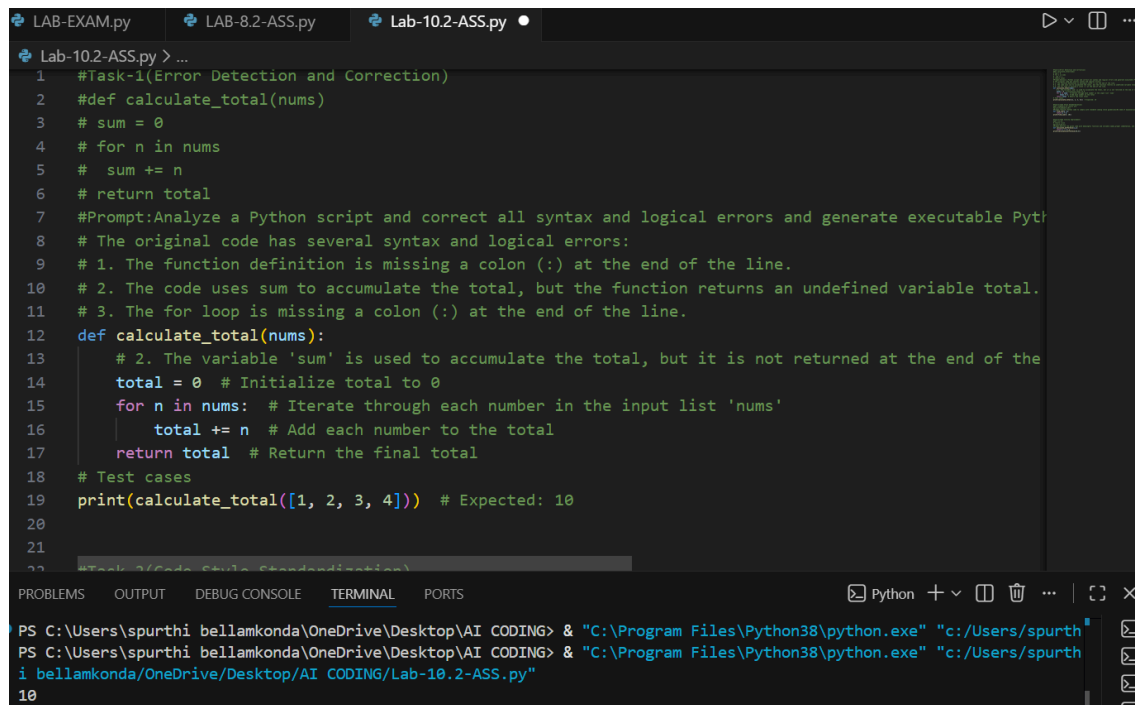
Use AI to analyze a Python script and correct all syntax and logical errors.

Sample Input Code:

```
def calculate_total(nums)
sum = 0
for n in nums
sum += n
return total
```

Expected Output-1:

Corrected and executable Python code with brief explanations of the identified syntax and logic errors.



The screenshot shows a VS Code editor with three tabs: LAB-EXAM.py, LAB-8.2-ASS.py, and Lab-10.2-ASS.py. The active tab is Lab-10.2-ASS.py, which contains a Python script. The script starts with a comment: `#Task-1(Error Detection and Correction)`. It defines a function `calculate_total(nums)` with several comments explaining its purpose and the errors it corrects. The function initializes `total = 0`, iterates through `nums` to calculate the sum, and returns `total`. Below the function, there are test cases: `print(calculate_total([1, 2, 3, 4]))` with a comment `# Expected: 10`. The terminal at the bottom shows the command to run the script: `PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-10.2-ASS.py"`, and the output is `10`.

```
1 #Task-1(Error Detection and Correction)
2 #def calculate_total(nums)
3 # sum = 0
4 # for n in nums
5 # sum += n
6 # return total
7 #Prompt:Analyze a Python script and correct all syntax and logical errors and generate executable Python code.
8 # The original code has several syntax and logical errors:
9 # 1. The function definition is missing a colon (:) at the end of the line.
10 # 2. The code uses sum to accumulate the total, but the function returns an undefined variable total.
11 # 3. The for loop is missing a colon (:) at the end of the line.
12 def calculate_total(nums):
13     # 2. The variable 'sum' is used to accumulate the total, but it is not returned at the end of the function.
14     total = 0 # Initialize total to 0
15     for n in nums: # Iterate through each number in the input list 'nums'
16         total += n # Add each number to the total
17     return total # Return the final total
18 # Test cases
19 print(calculate_total([1, 2, 3, 4])) # Expected: 10
20
21
22 #Task-2(Code Style Standardization)
```

```
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-10.2-ASS.py"
10
```

Task Description -2(Code Style Standardization)

Task:

Use AI to refactor Python code to comply with standard coding style guidelines.

Sample Input Code:

```
def findSum(a,b):return a+b
```

```
print(findSum(5,10))
```

Expected Output-2:

Well-structured, consistently formatted Python code following standard style conventions.

```
20
21
22 #Task-2(Code Style Standardization)
23 #def findSum(a,b):return a+b
24 #print(findSum(5,10))
25 #Prompt:refactor Python code to comply with standard coding style guidelines(No need of explanation)
26 def find_sum(a, b):
27     return a + b
28 print(find_sum(5, 10))
29
30
31 #Task-3(Code Clarity Improvement)
32 #def f(x,y):
33 # return x-y*2
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda\OneDrive\Desktop\AI CODING/Lab-10.2-ASS.py"

Task Description -3(Code Clarity Improvement)

Task:

Use AI to improve code readability without changing its functionality.

Sample Input Code:

```
def f(x,y):
return x-y*2
print(f(10,3))
```

Expected Output-3:

Python code rewritten with meaningful function and variable names, proper indentation, and improved clarity.

The screenshot shows a code editor with three tabs: LAB-EXAM.py, LAB-8.2-ASS.py, and Lab-10.2-ASS.py. The active tab is Lab-10.2-ASS.py, which contains the following Python code:

```
40 #Task-4(Structural Refactoring)
41 #print("Hello Ram")
42 #print("Hello Sita")
43 #print("Hello Ravi")
44 #Prompt:Generate a Modular Python code using reusable functions to eliminate repetition with proper doc
45 def greet(name):
46     """
47     Print a greeting message for the given name.
48
49     Parameters:
50     name (str): The name of the person to greet.
51
52     Returns:
53     None
54     """
55     print(f"Hello {name}")
56 # Example usage
57 greet("Ram") # Expected output: "Hello Ram"
58 greet("Sita") # Expected output: "Hello Sita"
59 greet("Ravi") # Expected output: "Hello Ravi"
60
```

The bottom panel of the editor shows the terminal output of the code execution:

```
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-10.2-ASS.py"
10
15
4
Hello Ram
Hello Sita
Hello Ravi
```

Task Description -5(Efficiency Enhancement)

Task:

Use AI to optimize Python code for better performance.

Sample Input Code:

```
numbers = [ ]
for i in range(1, 500000):
    numbers.append(i * i)
print(len(numbers))
```

Expected Output-5:

Optimized Python code that achieves the same result with improved performance.

The image shows a VS Code editor window with a Python file named 'Lab-10.2-ASS.py'. The code defines a function 'generate_squares(n)' that generates a list of squares from 1 to n-1. The function uses list comprehension for efficiency. An example usage is provided, generating squares up to 500,000. The output of the script is displayed in the 'TERMINAL' panel at the bottom, showing 'Hello Sita', 'Hello Ravi', and the number '499999'. The terminal prompt indicates the current directory is 'C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING'.

```
Lab-10.2-ASS.py > generate_squares
61 #Task-5(Efficiency Enhancement)
62 #numbers = [ ]
63 #for i in range(1, 500000):
64 # numbers.append(i * i)
65 #print(len(numbers))
66 #Prompt:Optimize the Python code that achieves the same result with improved performance with proper
67 def generate_squares(n):
68     """
69     Generate a list of squares of numbers from 1 to n-1.
70     Parameters:
71     n (int): The upper limit (exclusive) for generating squares.
72     Returns:
73     list: A list containing the squares of numbers from 1 to n-1.
74     Example:
75     >>> generate_squares(10)
76     [1, 4, 9, 16, 25, 36, 49, 64, 81]
77     """
78     return [i * i for i in range(1, n)] # Using list comprehension for efficient generation of squares
79 # Example usage
80 squares = generate_squares(500000)
81 print(len(squares)) # Expected output: 499999
82
83
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Hello Sita
Hello Ravi
499999
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>